Renao Yan

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Education

University of Washington, Seattle, WA, USA — Sep. 2024 – Present

Ph.D. Student (Research Assistant). Supervisor: Jonathan T.C. Liu.
Major: Mechanical Engineering.
GPA: 3.96 / 4.

Tsinghua University, China — Sep. 2021 – Jun. 2024

Master of Science. Supervisor: Tian GUAN.
Major: Electronic Information.
GPA: 3.62 / 4.

The Hong Kong University of Science and Technology, China — Jul. 2023 – Dec. 2023.

Research Assistant. Supervisor: Hao CHEN.

Hunan University, China — Sep. 2017 – Jun. 2021

Bachelor of Engineering. Supervisor: Qinghui HONG.
Major: Communication Engineering.
GPA: 3.76 / 4 (ranking: 1 / 121).

Experience

Ph.D. Student PERIOD

Project I: Handcraft-Guided Deep Learning Triage for Risk Stratification — Feb. 2025 – Present

Brief Introduction: While multiclass classification provides richer diagnostic guidance for pathologists than binary classification, existing deep learning-based triage frameworks often underperform in complex multiclass settings. To address this challenge, we developed a handcraft-guided triage framework for open-top light-sheet (OTLS) 3D pathology. Our approach incorporates segmentation masks, and fully leverages volumetric context across the entire 3D dataset to prioritize diagnostically relevant 2D cross-sections for expert review.
Result: Our method substantially improves the AUC in multiclass disease detection tasks, outperforming existing 2D and 2.5D deep learning baselines on prostate cancer and Barrett's esophagus datasets. By effectively highlighting high-risk tissue regions, the framework enhances diagnostic sensitivity while reducing the burden of manual review for pathologists.

Project II: Deep Learning Triage for 3D Pathology Risk Stratification — Sep. 2024 – Jun. 2025

Brief Introduction: Traditional 2D slide-based histopathology severely undersamples spatially heterogeneous tissues, with each thin section representing less than 1% of the specimen. To address this limitation, we developed a triage framework for open-top light-sheet (OTLS) 3D pathology, leveraging deep learning to identify diagnostically high-risk 2D cross sections from large 3D volumes. Our model integrates contextual information from neighboring depth levels to generate risk scores, prioritizing slices for expert review.
Result: Our framework significantly improves the sensitivity for detecting high-risk diseases, such as prostate cancer and Barrett's esophagus dysplasia, compared to conventional 2D histology. By efficiently directing pathologist attention to high-risk regions, the method enhances diagnostic accuracy while optimizing clinical workload.

MASTER PERIOD

Project III: Cooperative Game-Theoretic AI for WSI Classification — Jul. 2023 – Dec. 2023

Cooperating Organization: *Smart Lab, UKUST*.
Brief Introduction: To enhance whole slide image classification, we introduce an instance importance score, derived from Shapley values, to refine multiple instance learning. This approach improves model interpretability and facilitates more precise localization of diagnostically relevant regions.
Result: Our methodology surpasses existing state-of-the-art techniques. Additionally, the Shapley value-based instance importance score offers valuable class-wise interpretability. Superior performance and interpretation help pathologists make more comprehensive and efficient diagnoses in clinical practice.

Project IV: Multi-Scale Deep Learning for Cerebral Tumor Diagnosis Jan. 2023 – Jun. 2023

Cooperating Organization: *The First Affiliated Hospital, Sun Yat-sen University*.

Brief Introduction: We designed a multi-scale multiple-instance learning framework for brain tumor classification, offering an interpretable and automated deep learning solution to aid neuropathologists in decision-making.

Result: We developed a new dataset consisting of 2,085 cases and 3,098 WSIs from six hospitals. Our multi-magnification deep learning approach enhances brain tumor classification, improving diagnostic precision for neurosurgeons and pathologists.

Project V: Virtual Staining via Deep Learning for Computational Pathology Jan. 2022 – Jun. 2022

Cooperating Organization: *The Third People's Hospital of Shenzhen*.

Brief Introduction: Traditional Masson trichrome staining is time-consuming and costly. We propose a deep learning-based virtual staining pipeline that transforms H&E-stained liver biopsy images into high-fidelity Masson trichrome images, enabling efficient and cost-effective histopathological analysis.

Result: Our virtual staining model generates high-fidelity Masson trichrome images, reducing staining costs while maintaining clinical-grade diagnostic accuracy.

UNDERGRADUATE PERIOD

Project VI: Memristive Neural Network Circuit for License Plate Detection Jan. 2021 – Jun. 2021

Brief Introduction: Based on the least mean square algorithm, we have designed a memristive online self-learning neuron circuit. This circuit implementation allows for the realization of single- and multi-layer neural networks, enabling their application in pattern recognition and license plate detection.

Project VII: Bionic Memristive Circuit Implementation Jun. 2020 – Dec. 2020

Brief Introduction: We proposed a bionic neuron circuit with biological non-associative learning ability. Designed according to the synaptic programming characteristics of the memristor, the circuit can be applied to simulate visual fatigue and automate exposure compensation in cameras.

Selected Publications

ARTICLE

1. **Co-First Author**. "Hierarchically optimized multiple instance learning with multi-magnification pathological images for cerebral tumor diagnosis." *IEEE Journal of Biomedical and Health Informatics (JCR Q1)* (2025).
2. **First Author**. "Shapley values-enabled progressive pseudo bag augmentation for whole-slide image classification." *IEEE Transactions on Medical Imaging (JCR Q1)* (2024).
3. Other Author. "An accurate prediction of the origin for bone metastatic cancer using deep learning on digital pathological images." *EBioMedicine (JCR Q1)* (2023).
4. **First Author**. "Unpaired virtual histological staining using prior-guided generative adversarial networks." *Computerized Medical Imaging and Graphics (JCR Q1)* (2023).
5. **First Author**. "Multilayer memristive neural network circuit based on online learning for license plate detection." *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (JCR Q2)* (2021).
6. Second Author. "Memristive circuit implementation of biological nonassociative learning mechanism and its applications." *IEEE Transactions on Biomedical Circuits and Systems (JCR Q1)*(2020).

CONFERENCE

7. Second Author. "HIQ: One-shot quantization for histopathological image classification." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (2024).
8. **Corresponding Author**. "TexQ: Zero-shot network quantization with texture feature distribution calibration." *Conference on Neural Information Processing Systems (NeurIPS)* (2023).
9. Second Author. "ADEQ: Adaptive diversity enhancement for zero-shot quantization." *The 2023 International Conference on Neural Information Processing (ICONIP)* (2023).
10. **First Author**. "DEST: Deep enhanced swin transformer toward better scoring for NAFLD." *Chinese Conference on Pattern Recognition and Computer Vision* (2022).

PATENT

11. Second Inventor. "Shapley values-enabled progressive pseudo bag augmentation for whole slide image classification." US 63/644511 (2025).
12. Other Inventor. "A method for virtual staining hematoxylin-eosin images into Masson trichrome images." CN 202210635864.5 (2022).

13. Second Inventor. ”A signal generating device and adjustment method with habituation and sensitization.” CN 111291879A (2020).

Honors & Awards

Tsinghua University Outstanding Graduate (top 2%), Award on Tsinghua University	Jun. 2024
1st Class Scholarship, Award on Tsinghua University	Oct. 2023
Excellent Paper of the Year, Award on the Electronic Information Major	Oct. 2022
1st Class Scholarship, Award on Tsinghua Shenzhen International Graduate School	Oct. 2022
Undergraduate Excellent Innovative Graduation Project, Award on Hunan University	Jun. 2021
1st Class Scholarship, Award on Hunan University	Oct. 2020
Honorable Mention, Award on the Interdisciplinary Contest in Modeling	Apr. 2020
1st Class Scholarship, Award on Hunan University	Oct. 2019
Merit Student, Award on Hunan University	Oct. 2019
3rd Prize of Mathematics Competition (Non-mathematics Majors), Award on Hunan University	Jun. 2018

Skills and Service

Expertise:

- Deep Learning for Medical Image Diagnosis
- AI Interpretability for Medical Image Analysis
- Virtual Staining of Histopathology Images

Reviewer:

- IEEE Transactions on Industrial Informatics
- IEEE International Conference on Multimedia & Expo
- Chinese Conference on Pattern Recognition and Computer Vision
- International Journal of Circuit Theory and Applications